

Booklet No.

Civil Engineering Paper II

Time Allowed : Three Hours

Maximum Marks : 200

Medium : English

Type of Paper : Conventional

QUESTION PAPER SPECIFIC INSTRUCTIONS

Please read each of the following instructions carefully before attempting questions :

1. There are **EIGHT** questions divided in two Sections, out of which **FIVE** are to be attempted.
2. Question no. 1 and 5 are compulsory. Out of the remaining questions, **THREE** are to be attempted choosing at least **ONE** question from each Sections.
3. The number of marks carried by a question/sub question is indicated against it.
4. Keep in mind the word limit indicated in the question if any.
5. Wherever option has been given, only the required number of responses in the serial order attempted shall be assessed. Unless struck off, attempt of a question shall be counted even if attempted partly. Excess responses shall not be assessed and shall be ignored.
6. Candidates are expected to answer all the sub-questions of a question together. If sub-question of a question is attempted elsewhere (after leaving a few page or after attempting another question) the later sub-question shall be overlooked.
7. Any page or portion of the page left blank in the Answer Booklet must be clearly struck off.
8. Unless otherwise mentioned, symbol and notation have their usual standard meanings. Assume suitable data, if necessary and indicate the same clearly.
9. Neat sketches may be drawn, wherever required.
10. The medium of answer should be mentioned on the answer book as claimed in the application and printed on admission card. The answers written in medium other than the authorized medium will not be assessed and no marks will be assigned to them.

Note - Candidates will be allowed to use Scientific (Non-programmable type) calculators.

SEAL

SECTION - AAnswer **any five** of the following :**(8x5=40)**

1. (a) What are different errors in theodolite work ? The plan of an old survey plotted to a scale of 10 m to 1 cm carried a note stating that 'the chain was 0.8 links (16 cm) too short'. It was also found that the plan has shrunk so that a line originally 10 cm long was 9.77 cm. The area of a plot on the available plan was found to be 58.2 sq.cm. What is the correct area of the plan in hectares ? **(4+4)**
 - (b) Write short notes on the followings. **(4+4)**
 - (i) Ready-mix concrete
 - (ii) Preventive measures for corrosion of steel
 - (c) What is importance of orientation in building planning ? Explain principles of planning the building. **(2+6)**
 - (d) State methods of dimensional analysis. Describe Buckingham's π method. **(2+6)**
 - (e) What is indicator diagram ? Write a note on air-vessels in reciprocating pumps. **(2+6)**
 - (f) State methods of Irrigation. Explain drip irrigation with neat sketch. **(2+6)**
 - (g) What is sounding in surveying ? Explain methods of locating soundings. **(2+6)**
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2. (a) What do you understand by compound curve and reverse curve ? Where they are generally used ? **(8+7)**
 The following reciprocal observations were made from two points P and Q.
 Horizontal distance between P and Q = 6996 m.
 Angle of elevation of Q at P = $1^{\circ}56'10''$
 Angle of depression of P at Q = $1^{\circ}56'52''$
 Height of signal at P = 4.07 m
 Height of signal at Q = 3.87 m
 Height of instrument at P = 1.27 m
 Height of instrument at Q = 1.48 m
 Find difference in level between P and Q and refraction correction. Take $R \sin 1'' = 30.88$ m.
 - (b) What are mechanical properties of materials ? Describe in brief application of Fibre Glass Reinforced Plastic (FRP) in building industry. **(5+10)**
 - (c) State the instruments used for Electromagnetic distance measurement (EDM). **(3+7)**
 Explain Geodimeter with neat sketch.
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3. (a) State and explain factors affecting building planning. A 400 mm thick brick wall carries a load of 300 kN per running meter at ground level. The angle of repose of the soil is $33^{\circ}41'$ and weight of soil is 16 kN/m^3 . If the safe load on the soil be taken as 200 kN/m^2 . Design suitable foundation. (Assume (i) 10 kN as weight of footing and concrete block, (ii) take the projection of concrete block $j = 150 \text{ mm}$, (iii) Foundation concrete 1 : 3 : 6 (M_{10})). (7+8)

- (b) What is hydraulic jump ? (5+10)

State mathematical expression for sequent depth and energy loss.

Determine the length of back water curve caused by an afflux of 2.0 m in a rectangular channel of width 40 m and depth 2.5 m. The slope of the bed is 1 in 11000. Take Manning's $N = 0.03$.

- (c) State factors governing selection of type of dam. Explain them in brief. (10)
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4. (a) What is pump ? Explain cavitation in centrifugal pump. (8+7)

The diameters of an impeller of centrifugal pump at inlet and outlet are 30 cm and 60 cm respectively. Determine the minimum starting speed of the pump if it works against a head of 30 m.

- (b) What do you understand by initial and final regime conditions ? Explain in brief. (7+8)

Design an irrigation channel in alluvial soil according to Lacey's Silt theory with the following data :

Full supply discharge – 10 cumec

Lacey's silt factor – 0.9

Side Slope of channel – $\frac{1}{2} (H) : 1 (V)$

- (c) Explain fire protection of buildings. (10)
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SECTION – B

- 5. Write short notes on **any five** of the following : (8x5=40)**
- (a) Determination of road length using concept of 'Maximum Utility System'. (8)
 - (b) Tunnel Linings and its importance. (8)
 - (c) Types of values and sinking fund. (8)
 - (d) Types of Evaporimeters. (8)
 - (e) Methods of population forecast for design of water supply scheme. (8)
 - (f) Control of air pollution by process changes. (8)
 - (g) Advantages and disadvantages of Incineration of refuse. (8)
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- 6. (a) A new National Highway passing through rolling terrain and has a (5x3=15)**
horizontal curve of radius equal to ruling minimum radius. Design the following geometric features :
- (i) Stopping sight distance
 - (ii) Design rate of superelevation for mixed traffic condition.
 - (iii) Length of transition curve based on rate of change of centrifugal acceleration.
- Assume all other data suitably as per IRC recommendations.
- (b) Explain in brief the following : (5x3=15)**
- (i) Transfer of tunnel centre line through shaft.
 - (ii) The sequence of operations carried out for blasting of rocks for tunnelling.
 - (iii) Types of soft grounds (classification of soft grounds)
- (c) (i) List the information which should customarily appear in an (5x2=10)**
advertisement for tenders.
- (ii) A concrete mixer was purchased for Rs. 50,000/-. Assume salvage value of Rs. 5,000/- after five years. Calculate the book value of each year by calculating depreciation by constant percentage method.
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7. (a) What is specification ? State the different purpose of specification. Write the detail specifications for the following items of works : Excavation and cement concrete flooring in proportion 1 : 2 : 4. (15)
- (b) With reference to water treatment process, define sedimentation. State the various assumptions made while design of horizontal flow sedimentation tank. Design a rectangular sedimentation tank using the following data : (15)
1. Population 1 lac
 2. Overflow rate 25 m/day
 3. Depth of tank 3.2 m
 4. Rate of water supply to be 200 liters per hour per day
- (c) Explain briefly the terms storage coefficient and specific capacity, used in groundwater flow studies. The discharge from a fully penetrating well operating under steady state in a confined aquifer of 30 m thickness is 2100 litres/minute. The drawdowns observed at two observation wells located at 15 m and 150 m from the well are 3.2 m and 0.28 m respectively. Determine the transmissibility and the permeability of the aquifer. (10)
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8. (a) Attempt the following : (5x3=15)
- (i) What are the sources of solid waste and what are it's constituents ?
 - (ii) What are the problems created by solid wastes in urban areas ?
 - (iii) Discuss in brief composting as a method of disposal of solid waste.
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(b) Explain in detail the following :

(5×3=15)

- (i) Storage zones of reservoir
- (ii) Fixation of reservoir capacity with the help of mass curve of inflow and outflow.
- (iii) Uses and limitations of unit hydrograph.

(c) Explain spot speed, running speed, space-mean speed, time mean speed. Three vehicles are traversing a 1 km segment of highway and the following observation is made :

(10)

Name of Vehicle	Length of Vehicle	Travel Time
A	16 m	1.2 min.
B	18 m	1.5 min.
C	20 m	1.7 min.

Calculate :

- (i) Space Mean speed in kmph.
- (ii) If the above three vehicles pass a 1 km mark at their individual speed. What is the time mean speed of the three vehicles ?

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